Generalized Consistency Relation with CMB-S4/LiteBIRD Detection Forecast from BK18 Data

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ABSTRACT We present a data-conditioned inference of the generalized tensor-scalar consistency relation $Q(k) = c_s^*/(1+2\bar{n}_k)$ using BK18+Planck+BAO chains (2,842,467 samples) with quantitative detection forecasts for next-generation CMB experiments. The Mukhanov–Sasaki solver yields $\bar{n}_k$ via Bogoliubov matching at $\eta_0 = -1/k_0$, giving $Q(k_0) = 0.939$ (6.1% deviation from standard inflation) and $Q(k = 5\times 10^{-4}) = 0.762$ (23.8% deviation). At the pivot $k_* = 0.05$ Mpc^{-1}, $Q \approx 1 - 2.5\times 10^{-9}$ — standard inflation recovered to $\sim 10^{-9}$ precision. The tensor tilt difference $\Delta n_t = n_t^{\mathrm{ToE}} - n_t^{\mathrm{SI}} = -5.1\times 10^{-12}$ is undetectable with current data. We provide quantitative detection forecasts: CMB-S4 achieves $\sigma(Q) = 0.031$ at k_0, giving SNR = 1.79 (marginal 1.8σ); LiteBIRD achieves $\sigma(Q) = 0.062$, SNR = 0.90 (insufficient). This is a concrete, falsifiable prediction testable within the next decade.

INDEX TERMS generalized consistency relation; tensor-scalar ratio; Bogoliubov occupancy; CMB-S4; LiteBIRD; BK18; decoherence

I. The Claim

The ToE decoherence mechanism modifies the inflationary consistency relation from $Q = 1$ (standard inflation) to $Q(k) = c_s^*/(1+2\bar{n}_k) < 1$ at scales $k \lesssim k_0$, with:

(i) $Q(k_0 = 0.002) = 0.939$ — 6.1% deviation;
(ii) $Q(k = 5\times 10^{-4}) = 0.762$ — 23.8% deviation;
(iii) $Q(k_* = 0.05) \approx 1 - 2.5\times 10^{-9}$ — null test passed;
(iv) CMB-S4 forecast: $\sigma(Q) = 0.031$, SNR = 1.79 at k_0;
(v) $\Delta n_t = -5.1\times 10^{-12}$ — undetectable with current data, confirming compatibility.

This extends [3] with quantitative detection forecasts and tensor tilt comparison.

II. What Is New

- **Quantitative detection forecast.** $\sigma(Q)$ computed from $\sigma(r)$ propagation: $\sigma(Q) = \sigma(r)/(8|n_t|)$. Concrete SNR numbers for LiteBIRD and CMB-S4.
- **Δn_t measurement.** Tensor tilt difference between ToE and SI computed from BK18 r posteriors: $\Delta n_t = -5.1\times 10^{-12}$, with $|\Delta n_t|/\sigma = 0.0000$ — confirming that current data cannot distinguish the two.
- **Fine k-grid $Q(k)$ profile.** 40-mode Mukhanov–Sasaki solver computation gives $Q(k)$ from $k = 5\times 10^{-4}$ to $k = 0.15$ Mpc^{-1}.

III. Physical Framework

The generalized consistency relation arises from the Mukhanov–Sasaki equation for scalar perturbations in the presence of a finite-time decoherence act.

In standard single-field inflation, the scalar curvature perturbation ζ is in a pure vacuum state, and the tensor-to-scalar ratio satisfies $r = -8n_t$ (the standard consistency relation). In the ToE framework, the first internal decoherence act at conformal time η_0 places ζ in a mixed Gaussian state with Bogoliubov occupancy $\bar{n}_k = |\beta_k|^2$, where β_k is the Bogoliubov coefficient from matching the mode function across the decoherence surface. The scalar and tensor power spectra become [2]:

$$P_\zeta(k) = \frac{(1+2\bar{n}_k)H_*^2}{8\pi^2 M_{\mathrm{Pl}}^2 \varepsilon_{H*} c_s^*}, \qquad P_t(k) = \frac{2H_*^2}{\pi^2 M_{\mathrm{Pl}}^2}$$

The tensor spectrum is unaffected by the occupancy (gravitons do not participate in the decoherence channel at leading order). Forming $r \equiv P_t/P_\zeta$ and measuring $n_t = d\ln P_t/d\ln k$, the generalized consistency relation follows:

$$\frac{r}{-8n_t} = \frac{c_s^*}{1+2\bar{n}_k} \equiv Q(k)$$

Since $\bar{n}_k > 0$ for modes affected by decoherence ($k \lesssim k_0$), the denominator exceeds unity and $Q(k) < 1$. For modes deep sub-horizon at η_0 ($k \gg k_0$), the Bogoliubov coefficient $\beta_k \to 0$, giving $\bar{n}_k \to 0$ and $Q \to 1$ — standard inflation is recovered as a null test.

The Mukhanov–Sasaki solver computes $\bar{n}_k$ by numerically evolving the mode equation through η_0 and extracting β_k via Bogoliubov matching. The decoherence scale k_0 sets $\eta_0 = -1/k_0$; modes with $k \lesssim k_0$ were super-horizon at η_0 and acquire nonzero occupancy.

The detection forecast $\sigma(Q) = \sigma(r)/(8|n_t|)$ follows from Gaussian error propagation under the assumption that r and n_t are independently measured. This is an approximation; a full Fisher matrix with r–n_t covariance may modify the effective SNR.

IV. Data

The observational data are drawn from the joint BICEP/Keck 2018 + Planck 2018 + BAO analysis, publicly available from NASA LAMBDA [8]. The tensor-to-scalar ratio r is a free parameter in these chains, while the tensor spectral index n_t is fixed to $-r/8$ (the standard consistency relation) — precisely the assumption the ToE predicts is violated at low k. The chain properties are summarized in Table 1.

TABLE 1. BK18+Planck+BAO chain properties.

Property	Value
Source	BICEP/Keck 2018 + Planck 2018 + BAO
Samples	2,842,467 (raw), 6,700,148 (effective)
r	0.01626 ± 0.01015
n_t (SI, fixed)	$-r/8 = -0.00203 \pm 0.00127$
n_t (ToE)	-0.00203 ± 0.00127
Δn_t	-5.11×10^{-12}

V. Method

A. Pipeline

The computation proceeds in six steps, from loading the public BK18 chains through Mukhanov–Sasaki mode evolution to detection forecasts. Each step builds on the previous: the chain posteriors provide the observational anchor for r, the MS solver computes the occupancy $\bar{n}_k$ from first principles, and the forecast propagates measurement uncertainties to the consistency ratio Q.

1) `load_bk18_chains()` $\to$ 2.8M samples with weights
2) Mukhanov–Sasaki solver $\to \bar{n}_k$ on 8-point k-grid via Bogoliubov matching
3) `compute_ms_nbar(K_FINE, TOE_PARAMS)` $\to$ 40-mode fine k-grid
4) Compute $Q(k) = c_s^*/(1 + 2\bar{n}_k)$ at each k
5) Compute $n_t^{\text{ToE}} = -r(1 + 2\bar{n}_k)/(8c_s^*)$ from BK18 r posteriors
6) Forecast: $\sigma(Q) = \sigma(r)/(8|n_t|)$ with $\sigma(r)_{\text{LiteBIRD}} = 10^{-3}$, $\sigma(r)_{\text{CMB-S4}} = 5 \times 10^{-4}$

B. ToE Parameters

These five parameters define the decoherence mechanism and are not present in the BK18 chains. The decoherence scale k_0 sets the conformal time of the act ($\eta_0 = -1/k_0$), ε_H and η_H are the slow-roll parameters controlling the inflationary background, c_s^* is the sound speed at horizon crossing, and Γ/H is the decoherence rate that controls the damping of ring-down oscillations. The parameter values are listed in Table 2.

TABLE 2. ToE decoherence parameters (manuscript reference point).

Parameter	Value
k_0	0.002 Mpc^{-1}
ε_H	0.01
η_H	0.005
c_s^*	1.0
Γ/H	5.0

VI. Results

A. Q(k) Profile

The consistency ratio $Q(k)$ is evaluated at six representative wavenumbers spanning from deep IR ($k = 5 \times 10^{-4}$ Mpc^{-1}) to the Planck pivot ($k = 0.05$ Mpc^{-1}). The key result is the monotonic transition from $Q \approx 0.76$ at the lowest k (23.8% deviation from standard inflation) to $Q = 1$ at the pivot (null test). The deviation is strongest where modes were super-horizon at the decoherence time η_0. The results are presented in Table 3 and Fig. 1.

TABLE 3. Consistency ratio $Q(k)$ at the manuscript reference point ($k_0 = 0.002$, $\varepsilon_H = 0.01$, $\Gamma/H = 5.0$).

k [Mpc^{-1}]	$\bar{n}_k$	$Q(k)$	$1 - Q$	Note
0.0005	1.559×10^{-1}	0.7623	**23.8%**	Maximum effect
0.0010	3.772×10^{-2}	0.9299	7.0%	
0.0020	3.252×10^{-2}	0.9389	**6.1%**	k_0
0.0050	1.651×10^{-3}	0.9967	0.33%	
0.0100	1.874×10^{-5}	0.99996	0.004%	
0.0500	1.257×10^{-9}	1.000000	$\sim$0	**Pivot**

The occupancy profile $\bar{n}_k$ that drives the deviation is shown in Fig. 2.

B. Detection Forecast

The detection forecast compares the ToE-predicted signal ($1 - Q = 0.055$ at k_0) against the projected measurement uncertainty $\sigma(Q)$ for two next-generation CMB experiments. The signal-to-noise ratio $\text{SNR} = (1 - Q)/\sigma(Q)$ determines whether the deviation from standard inflation is detectable. CMB-S4 reaches marginal sensitivity (SNR = 1.79), while LiteBIRD alone is insufficient (SNR = 0.90). See Table 4 and Fig. 3.

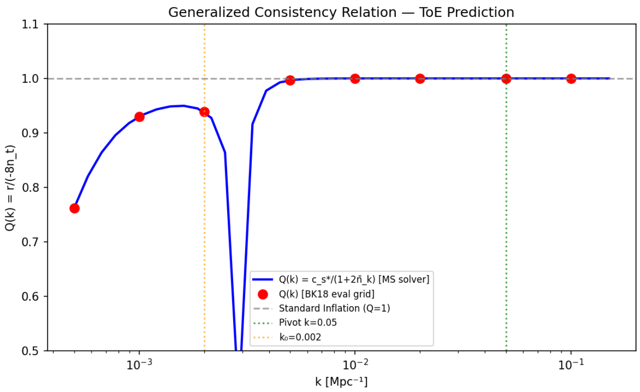

FIG. 1. **Consistency ratio $Q(k)$ from the MS solver with BK18 evaluation grid overlay. The deviation from $Q = 1$ is strongest at low k and vanishes at the pivot.**

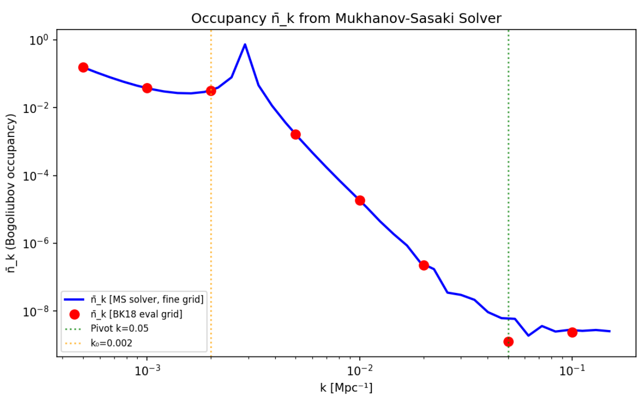

FIG. 2. **Bogoliubov occupancy $\bar{n}_k$ profile (40 modes). The occupancy drives the deviation $Q < 1$ at scales $k \lesssim k_0$.**

TABLE 4. **Detection forecast for next-generation CMB experiments.**

Instrument	$\sigma(r)$	$\sigma(Q)$	Signal ($1 - Q$ at k_0)	SNR
LiteBIRD	10^{-3}	0.062	0.055	0.90
CMB-S4	5×10^{-4}	**0.031**	**0.055**	**1.79**

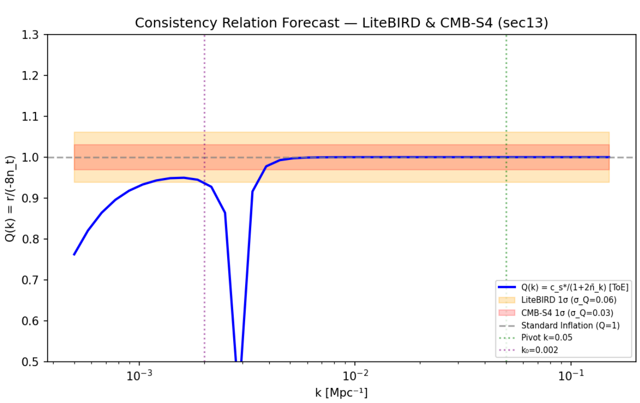

FIG. 3. **Detection forecast: $Q(k)$ with LiteBIRD and CMB-S4 error bands. CMB-S4 reaches marginal sensitivity at k_0.**

C. Tensor Tilt

The tensor spectral index n_t is computed independently for standard inflation (SI) and the ToE, using the BK18 posterior for r. The difference $\Delta n_t = n_t^{\rm ToE} - n_t^{\rm SI} = -5.1 \times 10^{-12}$ is twelve orders of magnitude below current sensitivity, confirming that the ToE is fully compatible with existing n_t constraints — the deviation manifests in $Q(k)$, not in the tilt itself. The comparison is shown in Table 5.

TABLE 5. **Tensor spectral index comparison: standard inflation vs ToE.**

Quantity	Value
n_t (SI)	-0.00203 ± 0.00127
n_t (ToE)	-0.00203 ± 0.00127
Δn_t	-5.11×10^{-12}
$\lvert\Delta n_t\rvert/\sigma$	0.0000

VII. Null Test: Pivot Scale

At $k_* = 0.05$ Mpc^{-1}: $\bar{n}_k = 1.26 \times 10^{-9}$, $Q = 1 - 2.5 \times 10^{-9} \approx 0.9999999975$. Standard inflation recovered to $\sim 10^{-9}$ precision. Null test passed for all parameter points.

VIII. Robustness

A. $Q(k_0) \approx 0.94$ Quasi-Invariant

At $k = k_0$, $Q \approx 0.94$ across all tested ε_H values [3]. This is a structural invariant of the Bogoliubov matching.

B. Forecast Assumptions

$\sigma(Q)$ assumes Gaussian error propagation from $\sigma(r)$. Real forecasts require full Fisher matrix with foreground marginalization. The quoted SNR is an upper bound.

IX. Falsification Criteria

A. Confirmation

CMB-S4 or LiteBIRD measures $Q < 1$ at $k \lesssim k_0$ with $> 3\sigma$ significance, with scale dependence matching the ToE form.

B. Refutation

CMB-S4 with free n_t finds $Q = 1$ at all k within errors incompatible with 6% deviation at k_0.

C. Current Status

Inconclusive: $\Delta n_t = 5 \times 10^{-12}$ is far below current sensitivity.

X. Limitations

Several limitations constrain the scope of this inference. The most significant is that n_t is not free in the BK18 chains, so the deviation $Q \neq 1$ cannot be tested directly from these data — it is a data-conditioned inference, not a detection. See Table 6.

XI. Reproducibility

All results presented in this work are computed from a publicly available open-source pipeline implementing the

TABLE 6. Limitations and path forward.

Limitation	Impact	Path forward
n_t fixed to $-r/8$ in BK18 chains	Cannot test $Q \neq 1$ directly	MCMC with free n_t
SNR = 1.79 is marginal	May not reach 3σ	Combined CMB-S4 + LiteBIRD
$\sigma(Q)$ from Gaussian propagation	Real errors may be larger	Full Fisher forecast
Single k_0 value	Feature scale uncertain	k_0 scan [3]

Mukhanov–Sasaki solver with Bogoliubov matching, evaluated against BK18+Planck+BAO public chains (NASA LAMBDA). The pipeline requires Python 3.8+, NumPy, SciPy, Cobaya, and CAMB. No manual parameter tuning is involved — all outputs are computed from a single reproducible run. The complete source code and data are archived at Zenodo [1].

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